POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2048 | Semester II | Lab | 1.5 Credits

Elements of Electrical & Electronics Engineering Lab

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics, communication, biomedical, microelectronics, instrumentation and related programmes listed in the official syllabus**
Contact hours	**45**
Teaching scheme	**0:0:3:0**
Credits	**1.5**
CIE / ESE	**Practical CIE 60 / Practical ESE 40**
Self-learning	**1.5 h/week; 22.5 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2048**, titled **Elements of Electrical & Electronics Engineering Lab**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

Source treatment: PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

Verified source note: No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

45
Contact hours

1.5
Credits

Practical CIE 60
CIE

Practical ESE 40
ESE

Course introduction

This laboratory course gives structured experience in domestic wiring, electrical measurements, regulator circuits and digital logic. It develops safe circuit construction, systematic observation, graph interpretation and professional record writing.

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Prerequisite foundation

Basic electrical circuits and passive/active components, concurrent with 2041 Elements of Electrical & Electronics Engineering.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
0	0	3	0	1.5 h/week; 22.5 h total	45	1.5	Practical CIE 60	Practical ESE 40

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Familiarise domestic wiring materials, standards and safety through hands-on circuit work.
2. Construct and verify electrical circuits and measure power/energy.
3. Test electronic components and regulator circuits.
4. Implement digital logic circuits and verify basic gates.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Identify wiring materials/standards and construct single-phase lighting circuits; verify voltage/current division.	Apply
CO2	Use measuring instruments for voltage, current, power and energy in single-phase circuits.	Apply
CO3	Test components, plot Zener characteristics and demonstrate fixed/adjustable regulators.	Apply
CO4	Construct basic amplifier/digital logic circuits and verify logic-gate operation.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Domestic Wiring and Circuit Division	Wiring-material familiarisation; single-phase lighting and staircase wiring; voltage/current division with lamps.	9 practical hours	CO1	Module 1
2	Power, Energy and Extension Board	Wattmeter use; energy-meter use; extension-board wiring.	9 practical hours	CO2	Module 2
3	Zener and Voltage Regulators	Zener line/load regulation; 78XX/79XX fixed regulators; LM317 adjustable regulator.	9 practical hours	CO3	Module 3
4	Digital Logic and Open-Ended Work	Digital trainer kit; diode/switch logic; NOT/AND/OR/NAND/NOR/ExOR/ExNOR verification; open-ended experiment.	9 practical hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Domestic Wiring and Circuit Division

Wiring-material familiarisation; single-phase lighting and staircase wiring; voltage/current division with lamps.

CO1 · 9 practical hours

Power, Energy and Extension Board

Wattmeter use; energy-meter use; extension-board wiring.

CO2 · 9 practical hours

Zener and Voltage Regulators

Zener line/load regulation; 78XX/79XX fixed regulators; LM317 adjustable regulator.

CO3 · 9 practical hours

Digital Logic and Open-Ended Work

Digital trainer kit; diode/switch logic; NOT/AND/OR/NAND/NOR/ExOR/ExNOR verification; open-ended experiment.

CO4 · 9 practical hours

MODULE 1

Domestic Wiring and Circuit Division

Wiring-material familiarisation; single-phase lighting and staircase wiring; voltage/current division with lamps.

9 practical hours | CO1 | Understand · Apply

Learning goals

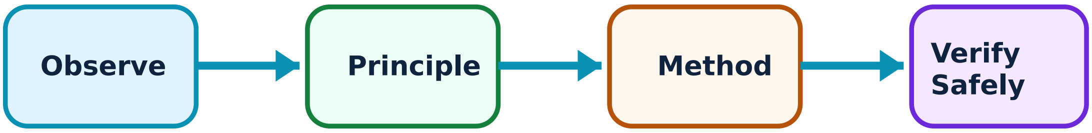
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Domestic Wiring and Circuit Division: identify the system, state its principle, follow the correct method and verify the result safely.

1. Wiring materials, standards and safety−

Domestic wiring requires suitable conductor, insulation, protection, switches, lamps and accessories. Familiarisation connects component identification to safe purpose and correct location in circuit.

Study and application method

Identify supply isolation point and phase/neutral/earth roles. Prepare circuit diagram and physical layout diagram before handling materials.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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Common mistake / precaution: Never energise a circuit until connections are inspected by the instructor.

2. Single-phase lighting and staircase control−

A lamp controlled by an SPST switch demonstrates basic lighting control. Two-way switches provide staircase control from two locations. The outcome is correct, safe construction and functional verification.

Study and application method

Wire one section at a time, check continuity with supply isolated, then test after approval. Use labels matching circuit diagram.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: Switching must follow safe phase-line practice; do not improvise terminal connections.

3. Voltage and current division with lamps−

In series, same current passes through each lamp and voltage divides; in parallel, branch voltage is common and current divides. Different lamp ratings make the practical effect visible, but observations need interpretation using circuit conditions.

Study and application method

Create a table for supply, individual lamp voltage/current and brightness observation, then compare it to division principle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: Brightness is not a substitute for measured electrical quantity; record instrument readings.

Energy-use calculator

Appliance power (W)

100

Hours used

5

Tariff per unit

6

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Power, Energy and Extension Board

Wattmeter use; energy-meter use; extension-board wiring.

9 practical hours · CO2 · Understand · Apply

Learning goals

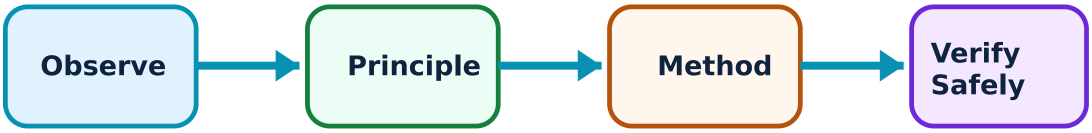
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Power, Energy and Extension Board: identify the system, state its principle, follow the correct method and verify the result safely.

1. Single-phase power measurement−

A wattmeter measures real power in an appropriate single-phase connection. Correct current-coil and potential-coil connection, range selection and reading are core practical skills.

Study and application method

Read nameplate/range, isolate before wiring, obtain instructor check and record voltage, current and power with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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Common mistake / precaution: Incorrect wattmeter connection can damage instrument; never guess terminal use.

2. Energy measurement and billing−

An energy meter records consumed electrical energy, commonly in kWh (units). The task relates meter reading change to a controlled single-phase load.

Study and application method

Record initial/final readings and elapsed time, then calculate consumption. Keep distinction clear: watt is power, kWh is energy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: Do not reset or tamper with installed meters; use laboratory arrangement only.

3. Extension-board assembly−

An extension board combines plug, cable, fuse, sockets and switches into a portable assembly. It must be mechanically sound and electrically safe.

Study and application method

Plan layout, strip conductors correctly, tighten terminals, provide strain relief and verify continuity/insulation under supervision.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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Common mistake / precaution: Loose terminals and damaged insulation are fire/shock hazards; reject unsafe assemblies.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Zener and Voltage Regulators

Zener line/load regulation; 78XX/79XX fixed regulators; LM317 adjustable regulator.

9 practical hours | **CO3** | **Understand · Apply**

Learning goals

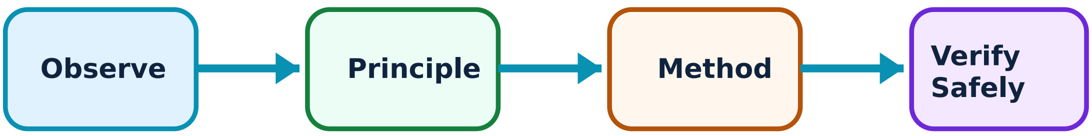
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Zener and Voltage Regulators: identify the system, state its principle, follow the correct method and verify the result safely.

1. Zener voltage regulator characteristics−

A Zener regulator uses reverse-breakdown behaviour with suitable series resistance to maintain approximately controlled output over operating range. Line regulation tests input variation; load regulation tests output change with load.

Study and application method

Start with device rating and series-resistor check. Vary one quantity at a time, record input/output and plot a labelled characteristic.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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Common mistake / precaution: Excess Zener current can exceed power rating; verify safe limits before test.

2. Fixed positive and negative regulator ICs−

78XX and 79XX families illustrate fixed positive and negative regulation. Skill includes pin identification, correct polarity, input/output measurement and awareness of operating conditions.

Study and application method

Confirm specified package pinout, use common reference correctly and measure no-load/load output after inspection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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Common mistake / precaution: Reversed input or wrong pin connection can destroy an IC; body appearance is not enough.

3. LM317 adjustable regulator−

LM317 demonstrates adjustable regulation and line/load characteristics. A resistance network establishes output setting within permitted limits.

Study and application method

Build circuit exactly as instructed, adjust gradually and log input/output/load changes. Present a graph/table supporting conclusion.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: Heat dissipation rises with voltage drop and load current; do not exceed ratings.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Digital Logic and Open-Ended Work

Digital trainer kit; diode/switch logic; NOT/AND/OR/NAND/NOR/ExOR/ExNOR verification; open-ended experiment.

9 practical hours · CO4 · Understand · Apply

Learning goals

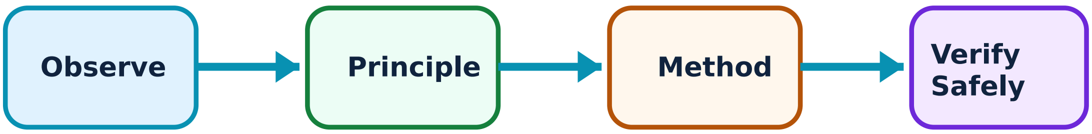
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Digital Logic and Open-Ended Work: identify the system, state its principle, follow the correct method and verify the result safely.

1. Digital trainer kit and discrete logic−

A trainer kit provides controlled logic supplies and input/output indicators. Discrete switches, diodes and resistors can demonstrate selected AND/OR functions. A circuit is successful only when output matches defined logic behaviour.

Study and application method

Prepare truth table before connection. Change one input combination at a time and record actual output alongside expected output.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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Common mistake / precaution: Avoid shorting supply rails and do not change wiring with power on unless approved procedure permits it.

2. Logic-gate truth-table verification−

NOT, AND, OR, NAND, NOR, ExOR and ExNOR gates are verified through complete input combinations. Skill is systematic testing, not memorising one output example.

Study and application method

Label inputs/outputs, use binary 0/1 consistently and test all rows. For two-input gates verify four combinations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: A floating input can give unreliable output; use defined logic levels.

3. Open-ended experiment−

The official open-ended activity asks students to conceive, design and implement an experiment related to course outcomes, possibly extending an experiment, integrating concepts or making a small prototype.

Study and application method

Write aim, constraints, circuit/flowchart, materials, safety plan, method, observations, analysis and conclusion. Retain test evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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Common mistake / precaution: Originality does not excuse unsafe design; obtain approval before energising prototype.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result

→ precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Domestic Wiring and Circuit Division

Use the concept flow to revise observation, principle, method and verification.

Module 2: Power, Energy and Extension Board

Use the concept flow to revise observation, principle, method and verification.

Module 3: Zener and Voltage Regulators

Use the concept flow to revise observation, principle, method and verification.

Module 4: Digital Logic and Open-Ended Work

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Analyse domestic phase-line switching, safety mechanisms, lamp ratings and domestic series/parallel use.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Record home-appliance power/energy observations within safe non-contact limits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare component, diode-testing and regulator-IC pinout charts.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Simulate NAND logic and document logic-family and diode-logic observations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	Regular attendance and punctuality.
Continuous evaluation	Practical CIE 60	Preparation, setup, observation, analysis, viva, record and discipline.
Practical tests	As stated in supplied PDF	Procedure/write-up, execution, result, viva and certified record.
Open-ended experiment	Where listed in supplied PDF	Originality, plan, execution/data, analysis/presentation and teamwork.
End-semester assessment	Practical ESE 40	Safe completion, record and viva according to official scheme.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Domestic Wiring and Circuit Division

Explain Wiring materials, standards and safety and state one correct method, application or precaution.−

Domestic wiring requires suitable conductor, insulation, protection, switches, lamps and accessories. Familiarisation connects component identification to safe purpose and correct location in circuit.

Answer framework: Identify supply isolation point and phase/neutral/earth roles. Prepare circuit diagram and physical layout diagram before handling materials.

Important point: Never energise a circuit until connections are inspected by the instructor.

Explain Single-phase lighting and staircase control and state one correct method, application or precaution.−

A lamp controlled by an SPST switch demonstrates basic lighting control. Two-way switches provide staircase control from two locations. The outcome is correct, safe construction and functional verification.

Answer framework: Wire one section at a time, check continuity with supply isolated, then test after approval. Use labels matching circuit diagram.

Important point: Switching must follow safe phase-line practice; do not improvise terminal connections.

Explain Voltage and current division with lamps and state one correct method, application or precaution.−

In series, same current passes through each lamp and voltage divides; in parallel, branch voltage is common and current divides. Different lamp ratings make the practical effect visible, but observations need interpretation using circuit conditions.

Answer framework: Create a table for supply, individual lamp voltage/current and brightness observation, then compare it to division principle.

Important point: Brightness is not a substitute for measured electrical quantity; record instrument readings.

Module 2 — Power, Energy and Extension Board

Explain Single-phase power measurement and state one correct method, application or precaution.−

A wattmeter measures real power in an appropriate single-phase connection. Correct current-coil and potential-coil connection, range selection and reading are core practical skills.

Answer framework: Read nameplate/range, isolate before wiring, obtain instructor check and record voltage, current and power with units.

Important point: Incorrect wattmeter connection can damage instrument; never guess terminal use.

Explain Energy measurement and billing and state one correct method, application or precaution.−

An energy meter records consumed electrical energy, commonly in kWh (units). The task relates meter reading change to a controlled single-phase load.

Answer framework: Record initial/final readings and elapsed time, then calculate consumption. Keep distinction clear: watt is power, kWh is energy.

Important point: Do not reset or tamper with installed meters; use laboratory arrangement only.

Explain Extension-board assembly and state one correct method, application or precaution.−

An extension board combines plug, cable, fuse, sockets and switches into a portable assembly. It must be mechanically sound and electrically safe.

Answer framework: Plan layout, strip conductors correctly, tighten terminals, provide strain relief and verify continuity/insulation under supervision.

Important point: Loose terminals and damaged insulation are fire/shock hazards; reject unsafe assemblies.

Module 3 — Zener and Voltage Regulators

Explain Zener voltage regulator characteristics and state one correct method, application or precaution.−

A Zener regulator uses reverse-breakdown behaviour with suitable series resistance to maintain approximately controlled output over operating range. Line regulation tests input variation; load regulation tests output change with load.

Answer framework: Start with device rating and series-resistor check. Vary one quantity at a time, record input/output and plot a labelled characteristic.

Important point: Excess Zener current can exceed power rating; verify safe limits before test.

Explain Fixed positive and negative regulator ICs and state one correct method, application or precaution.−

78XX and 79XX families illustrate fixed positive and negative regulation. Skill includes pin identification, correct polarity, input/output measurement and awareness of operating conditions.

Answer framework: Confirm specified package pinout, use common reference correctly and measure no-load/load output after inspection.

Important point: Reversed input or wrong pin connection can destroy an IC; body appearance is not enough.

Explain LM317 adjustable regulator and state one correct method, application or precaution.−

LM317 demonstrates adjustable regulation and line/load characteristics. A resistance network establishes output setting within permitted limits.

Answer framework: Build circuit exactly as instructed, adjust gradually and log input/output/load changes. Present a graph/table supporting conclusion.

Important point: Heat dissipation rises with voltage drop and load current; do not exceed ratings.

Module 4 — Digital Logic and Open-Ended Work

Explain Digital trainer kit and discrete logic and state one correct method, application or precaution.−

A trainer kit provides controlled logic supplies and input/output indicators. Discrete switches, diodes and resistors can demonstrate selected AND/OR functions. A circuit is successful only when output matches defined logic behaviour.

Answer framework: Prepare truth table before connection. Change one input combination at a time and record actual output alongside expected output.

Important point: Avoid shorting supply rails and do not change wiring with power on unless approved procedure permits it.

Explain Logic-gate truth-table verification and state one correct method, application or precaution.−

NOT, AND, OR, NAND, NOR, ExOR and ExNOR gates are verified through complete input combinations. Skill is systematic testing, not memorising one output example.

Answer framework: Label inputs/outputs, use binary 0/1 consistently and test all rows. For two-input gates verify four combinations.

Important point: A floating input can give unreliable output; use defined logic levels.

Explain Open-ended experiment and state one correct method, application or precaution.−

The official open-ended activity asks students to conceive, design and implement an experiment related to course outcomes, possibly extending an experiment, integrating concepts or making a small prototype.

Answer framework: Write aim, constraints, circuit/flowchart, materials, safety plan, method, observations, analysis and conclusion. Retain test evidence.

Important point: Originality does not excuse unsafe design; obtain approval before energising prototype.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Domestic Wiring and Circuit Division.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Power, Energy and Extension Board.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Zener and Voltage Regulators.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Digital Logic and Open-Ended Work.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Wiring-material familiarisation; single-phase lighting and staircase wiring; voltage/current division with lamps..
2. Develop a complete answer or practical plan for: Wattmeter use; energy-meter use; extension-board wiring..
3. Develop a complete answer or practical plan for: Zener line/load regulation; 78XX/79XX fixed regulators; LM317 adjustable regulator..
4. Develop a complete answer or practical plan for: Digital trainer kit; diode/switch logic; NOT/AND/OR/NAND/NOR/ExOR/ExNOR verification; open-ended experiment..

LAST-MILE REVISION

Quick Revision

Module 1: Domestic Wiring and Circuit Division

Wiring-material familiarisation; single-phase lighting and staircase wiring; voltage/current division with lamps.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Power, Energy and Extension Board

Wattmeter use; energy-meter use; extension-board wiring.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Zener and Voltage Regulators

Zener line/load regulation; 78XX/79XX fixed regulators; LM317 adjustable regulator.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Digital Logic and Open-Ended Work

Digital trainer kit; diode/switch logic; NOT/AND/OR/NAND/NOR/ExOR/ExNOR verification; open-ended experiment.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Wiring materials, standards and safety	Domestic wiring requires suitable conductor, insulation, protection, switches, lamps and accessories.
Single-phase lighting and staircase control	A lamp controlled by an SPST switch demonstrates basic lighting control.
Voltage and current division with lamps	In series, same current passes through each lamp and voltage divides; in parallel, branch voltage is common and current divides.
Single-phase power measurement	A wattmeter measures real power in an appropriate single-phase connection.
Energy measurement and billing	An energy meter records consumed electrical energy, commonly in kWh (units).
Extension-board assembly	An extension board combines plug, cable, fuse, sockets and switches into a portable assembly.
Zener voltage regulator characteristics	A Zener regulator uses reverse-breakdown behaviour with suitable series resistance to maintain approximately controlled output over operating range.
Fixed positive and negative regulator ICs	78XX and 79XX families illustrate fixed positive and negative regulation.
LM317 adjustable regulator	LM317 demonstrates adjustable regulation and line/load characteristics.
Digital trainer kit and discrete logic	A trainer kit provides controlled logic supplies and input/output indicators.
Logic-gate truth-table verification	NOT, AND, OR, NAND, NOR, ExOR and ExNOR gates are verified through complete input combinations.
Open-ended experiment	The official open-ended activity asks students to conceive, design and implement an experiment related to course outcomes, possibly extending an experiment, integrating concepts or making a small prototype.

LEARNING RESOURCES

References

Officially listed print resources

1. Paul B. Zbar, Albert P. Malvino and Michael A. Miller, Basic Electronics: A Text-Lab Manual.
2. K. A. Navas, Electronics Lab Manuals.
3. Syammohan S. and Kuriachan D., Electronics Lab Manuals.

Officially listed web resources

- https://www.digimat.in/nptel/courses/video/108105112/L01.html
- https://www.digimat.in/nptel/courses/video/108101091/L01.html

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